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Cultivation of food plants for the silkworms

The first component, is to grow the food plants for the silkworms. Mulberry leaves are widely used as food for silkworm *Bombyx mori* and the cultivation of mulberry is called as **Moriculture**. Presently improved mulberry varieties like Victory1, S36, G2 and G4 which can withstand various agro - climatic and soil conditions are used for planting. The favourable season for cultivating of the mulberry plants is June, July, November and December. The mulberry crop production technology includes land preparation, preparation of cutting, planting techniques, maintenance of mulberry nursery, disease and pest management and uprooting for raising new mulberry gardens. Mulberry is also being grown as tree plant at an height of 123-152 cm with 20 x 20 cm or 25 x 25 cm spacing to harvest better silkworm cocoon crops.

Rearing of silkworms

The second component is the rearing of silkworm. A typical rearing house (6m x 4m x 3.5m) is constructed on an elevated place under shade to accommodate 100 dfls (disease free layings). Space of 1m should be provided surrounding the rearing house. Sufficient windows and ventilators should be provided for free circulation of air inside the rearing house. The windows and ventilators should be covered with nylon net to restrict the entry of uzi flies and other insects. Apart from the specified area of the rearing house; the following appliances such as hygrometer, power sprayers, rearing stands, foam pads, wax coated paraffin papers, nylon nets, baskets for keeping leaves, gunny bags, rotary or bamboo mountages and drier are needed for effective rearing of silkworms. The steps involved in rearing process of silkworm are disinfection of rearing house, incubation of eggs, brushing, young larval rearing and late age larval rearing.

The selected healthy silk moths are allowed to mate for 4 hours. Female moth is then kept in a dark plastic bed, it lays about 400 eggs in 24 hours; the female is taken out, crushed and examined for any disease, only certified disease-free eggs are reared for industrial purpose. The eggs are incubated in an incubator. The small larvae (caterpillars) hatch between 7-10 days. These larvae are kept in trays inside a rearing house at a temperature of about 20°C - 25°C. These are first fed on chopped mulberry leaves. After 4-5 days fresh leaves are provided. As the larvae grow, they are transferred to fresh leaves on clean trays, when fully grown they spin cocoons. Their maturity is achieved in about 45 days. At this stage the salivary glands (silk glands) starts secreting silk to spin cocoons.

Post cocoon processing

The method of obtaining silk thread from the cocoon is known as post cocoon processing. This includes **stifling** and **reeling.**

The process of killing the cocoons is called stifling. The process of removing the threads from the killed cocoon is called reeling. For reeling silk the cocoons are gathered about 8 -10 days after spinning had begun. The cocoons are first treated by steam or dry heat to kill the insect inside. This is necessary to prevent the destruction of the continuous fibre by the emergence of the moth. The cocoons are